Artificial Intelligence & Machine Learning

A Technical Reference Document for RAG Evaluation

This document serves as a structured knowledge base covering core concepts, algorithms, architectures, and milestones in Artificial Intelligence (AI) and Machine Learning (ML). Each section contains precise factual statements suitable for retrieval-augmented generation (RAG) system testing.

1. Foundations of Artificial Intelligence

1.1 Definition and Scope

Artificial Intelligence is the branch of computer science concerned with building systems that can perform tasks that typically require human intelligence. These tasks include reasoning, learning, perception, natural language understanding, and decision-making.

- The term 'Artificial Intelligence' was coined by John McCarthy in 1956 at the Dartmouth Conference.
- Alan Turing proposed the Turing Test in 1950 in his paper 'Computing Machinery and Intelligence'.
- The Turing Test evaluates a machine's ability to exhibit intelligent behaviour indistinguishable from a human.
- AI is broadly categorised into Narrow AI (ANI), General AI (AGI), and Super AI (ASI).
- Narrow AI is designed for a specific task, such as image recognition or playing chess.

1.2 History and Key Milestones

The history of AI spans over seven decades and includes several periods of rapid progress and setbacks known as 'AI winters'.

- The first AI program, Logic Theorist, was created by Allen Newell and Herbert A. Simon in 1955.
- ELIZA, an early natural language processing program, was developed at MIT by Joseph Weizenbaum between 1964 and 1966.
- The first AI winter occurred approximately between 1974 and 1980 due to a lack of computational resources and unmet expectations.
- The second AI winter lasted from roughly 1987 to 1993.
- IBM's Deep Blue defeated world chess champion Garry Kasparov in 1997.
- IBM's Watson won the quiz show Jeopardy! in February 2011, defeating human champions Ken Jennings and Brad Rutter.
- AlphaGo, developed by DeepMind, defeated professional Go player Lee Sedol 4-1 in March 2016.

2. Machine Learning

2.1 Definition and Types

Machine Learning is a subset of AI that enables systems to learn and improve from experience without being explicitly programmed. ML algorithms build a model based on sample data (training data) to make predictions or decisions.

- Machine Learning was formally defined by Arthur Samuel in 1959 as a field of study that gives computers the ability to learn without being explicitly programmed.
- The three primary types of machine learning are: Supervised Learning, Unsupervised Learning, and Reinforcement Learning.
- In Supervised Learning, the model is trained on labelled data where input-output pairs are provided.
- In Unsupervised Learning, the model finds patterns in unlabelled data without explicit guidance.
- In Reinforcement Learning, an agent learns by interacting with an environment and receiving rewards or penalties.
- Semi-supervised Learning uses a small amount of labelled data combined with a large amount of unlabelled data.
- Self-supervised Learning generates labels from the input data itself, without human annotation.

2.2 Key Algorithms

Machine learning encompasses a wide variety of algorithms, each suited to specific problem types.

- Linear Regression is used to model the relationship between a continuous dependent variable and one or more independent variables.
- Logistic Regression is a classification algorithm that predicts the probability of a binary outcome.
- A Decision Tree splits data into branches based on feature values to make predictions.
- Random Forest is an ensemble method that combines multiple decision trees to improve accuracy and reduce overfitting.
- Support Vector Machines (SVM) find the optimal hyperplane that maximises the margin between classes.
- K-Nearest Neighbours (KNN) classifies a data point based on the majority class of its k nearest neighbours.
- K-Means Clustering partitions data into k clusters by minimising intra-cluster variance.
- Naive Bayes is a probabilistic classifier based on Bayes' theorem, assuming feature independence.
- Gradient Boosting builds an ensemble of weak learners sequentially, with each learner correcting the errors of the previous one.

- XGBoost (Extreme Gradient Boosting) was introduced by Tianqi Chen in 2014 and is known for its speed and performance.

2.3 Model Evaluation Metrics

Evaluating the performance of ML models requires domain-appropriate metrics.

- Accuracy is the ratio of correctly predicted instances to the total instances.
- Precision measures the proportion of true positives among all positive predictions.
- Recall (Sensitivity) measures the proportion of actual positives correctly identified by the model.
- F1 Score is the harmonic mean of Precision and Recall.
- ROC-AUC (Area Under the Receiver Operating Characteristic Curve) evaluates classification model performance across all thresholds.
- Mean Absolute Error (MAE) measures the average magnitude of errors in regression predictions.
- Mean Squared Error (MSE) penalises larger errors more heavily than MAE due to squaring.
- R-squared (R2) indicates the proportion of variance in the dependent variable explained by the model.

3. Deep Learning

3.1 Neural Networks

Deep Learning is a subfield of machine learning based on artificial neural networks with multiple layers. These networks automatically learn hierarchical representations of data.

- An artificial neural network (ANN) is inspired by the biological neural networks in the human brain.
- A perceptron, introduced by Frank Rosenblatt in 1958, is the simplest form of a neural network.
- The backpropagation algorithm, used to train neural networks, was popularised by Rumelhart, Hinton, and Williams in 1986.
- A deep neural network typically has more than two hidden layers between the input and output layers.
- Activation functions introduce non-linearity into neural networks. Common ones include ReLU, Sigmoid, Tanh, and Softmax.
- ReLU (Rectified Linear Unit) outputs the input if positive and zero otherwise, and is the most widely used activation function.
- The vanishing gradient problem occurs when gradients become extremely small during backpropagation, hampering learning in deep networks.
- Dropout is a regularisation technique where random neurons are deactivated during training to prevent overfitting.
- Batch Normalisation normalises the inputs to each layer, accelerating training and improving stability.

3.2 Convolutional Neural Networks (CNNs)

CNNs are specialised neural networks designed for processing structured grid data such as images.

- CNNs use convolutional layers to automatically learn spatial hierarchies of features from input images.
- AlexNet, developed by Alex Krizhevsky, won the ImageNet competition in 2012 with a top-5 error rate of 15.3%.
- VGGNet, introduced by the Visual Geometry Group at Oxford, demonstrated that network depth is a critical factor in performance.
- ResNet (Residual Network), introduced by Microsoft Research in 2015, uses skip connections to enable training of very deep networks (up to 152 layers).
- Pooling layers in CNNs reduce spatial dimensions and help achieve translational invariance.
- Max Pooling selects the maximum value in each pooling window.

3.3 Recurrent Neural Networks (RNNs)

RNNs are designed for sequential data by maintaining a hidden state that captures information from previous time steps.

- RNNs suffer from the vanishing gradient problem, making it difficult to learn long-term dependencies.
- LSTM (Long Short-Term Memory), introduced by Hochreiter and Schmidhuber in 1997, addresses the vanishing gradient problem using gating mechanisms.
- An LSTM cell contains three gates: the input gate, forget gate, and output gate.
- GRU (Gated Recurrent Unit), proposed by Cho et al. in 2014, is a simplified version of LSTM with two gates: reset and update.
- Bidirectional RNNs process sequences in both forward and backward directions.

3.4 Transformers and Attention

The Transformer architecture revolutionised deep learning, particularly in natural language processing.

- The Transformer architecture was introduced in the paper 'Attention Is All You Need' by Vaswani et al. in 2017.
- The key innovation of Transformers is the self-attention mechanism, which allows the model to weigh the importance of different tokens in a sequence.
- Multi-head attention applies several attention functions in parallel, allowing the model to jointly attend to information from different representation subspaces.
- BERT (Bidirectional Encoder Representations from Transformers) was introduced by Google in 2018 and uses bidirectional training of Transformers.
- GPT (Generative Pre-trained Transformer) was introduced by OpenAI. GPT-1 was released in 2018, GPT-2 in 2019, and GPT-3 in 2020.
- GPT-3 contains 175 billion parameters and was trained on approximately 570 GB of text data.

- GPT-4 was released by OpenAI in March 2023.
- Positional encoding in Transformers injects information about the position of tokens in the sequence.

4. Natural Language Processing (NLP)

4.1 Core NLP Tasks

Natural Language Processing is a field at the intersection of AI and linguistics concerned with the interaction between computers and human language.

- Tokenisation is the process of splitting text into individual tokens such as words or subwords.
- Named Entity Recognition (NER) identifies and classifies named entities in text into categories such as person, organisation, and location.
- Part-of-Speech (POS) tagging assigns grammatical labels (noun, verb, adjective) to each token.
- Sentiment Analysis determines the emotional tone (positive, negative, neutral) expressed in text.
- Machine Translation automatically converts text from one language to another.
- Text Summarisation generates a shorter version of a text while preserving key information.
- Question Answering systems locate or generate answers to questions posed in natural language.

4.2 Word Embeddings

Word embeddings represent words as dense, low-dimensional vectors that capture semantic relationships.

- Word2Vec, developed by Mikolov et al. at Google in 2013, learns word embeddings using two architectures: CBOW and Skip-gram.
- GloVe (Global Vectors for Word Representation) was introduced by Stanford researchers in 2014.
- FastText, developed by Facebook AI Research, represents words as bags of character n-grams.
- Contextual embeddings, such as those from BERT and ELMo, produce different vector representations for the same word depending on context.
- ELMo (Embeddings from Language Models) was introduced by Allen Institute for AI in 2018.

5. Computer Vision

Computer Vision is an interdisciplinary field that enables computers to interpret and understand visual information from images and videos.

- Object Detection identifies and locates objects within an image, outputting bounding boxes and class labels.

- YOLO (You Only Look Once), introduced in 2016 by Redmon et al., performs object detection in real time using a single neural network pass.
- Image Segmentation assigns a class label to every pixel in an image.
- Semantic Segmentation classifies each pixel into a category without distinguishing between instances.
- Instance Segmentation distinguishes between separate instances of the same object class.
- Mask R-CNN, introduced by He et al. in 2017, extends Faster R-CNN to perform instance segmentation.
- Generative Adversarial Networks (GANs), introduced by Ian Goodfellow in 2014, consist of a generator and a discriminator trained adversarially.
- Diffusion Models, the basis for modern image generation systems like DALL-E 2 and Stable Diffusion, generate images by reversing a gradual noising process.
- CLIP (Contrastive Language-Image Pretraining), developed by OpenAI and released in January 2021, learns visual concepts from natural language supervision.

6. Reinforcement Learning

Reinforcement Learning (RL) is a paradigm where an agent learns to maximise cumulative reward through interactions with an environment.

- The key components of RL are: Agent, Environment, State, Action, Reward, and Policy.
- The Markov Decision Process (MDP) is the mathematical framework for modelling RL problems.
- Q-Learning is a model-free RL algorithm that learns the value of actions in given states.
- Deep Q-Network (DQN), introduced by DeepMind in 2015, combines Q-Learning with deep neural networks to play Atari games at superhuman level.
- Policy Gradient methods directly optimise the policy by computing gradients of expected rewards.
- Proximal Policy Optimisation (PPO), introduced by OpenAI in 2017, is a stable and widely used policy gradient method.
- AlphaStar, developed by DeepMind, reached Grandmaster level in StarCraft II using a combination of supervised learning and RL.
- OpenAI Five defeated professional Dota 2 players in 2019 using RL trained through self-play.

7. AI Ethics, Safety, and Governance

As AI systems become more capable and widespread, questions of ethics, safety, fairness, and governance have gained increasing importance.

- AI Bias refers to systematic and unfair discrimination in AI outputs, often stemming from biased training data.

■ The EU Artificial Intelligence Act, proposed in April 2021, is the world's first comprehensive legal framework for AI regulation.

■ Explainable AI (XAI) refers to methods and techniques that make AI decisions interpretable and understandable to humans.

■ LIME (Local Interpretable Model-agnostic Explanations) explains individual predictions of any classifier.

■ SHAP (SHapley Additive exPlanations) uses game-theoretic Shapley values to attribute predictions to features.

■ Differential Privacy is a technique that adds calibrated noise to data or model outputs to protect individual privacy.

■ Federated Learning allows models to be trained across decentralised devices without sharing raw data.

■ AI Alignment is the research area focused on ensuring AI systems act in accordance with human values and intentions.

■ Anthropic was founded in 2021 by Dario Amodei, Daniela Amodei, and other former OpenAI members, with a focus on AI safety.

8. Notable Models, Frameworks, and Tools

8.1 Deep Learning Frameworks

■ TensorFlow, developed by Google Brain, was open-sourced in November 2015.

■ PyTorch, developed by Facebook's AI Research lab, was released in 2016 and is widely used in research.

■ Keras is a high-level deep learning API, now integrated as tf.keras within TensorFlow.

■ JAX, developed by Google, supports automatic differentiation and GPU/TPU acceleration for numerical computing.

8.2 Large Language Models (LLMs)

Large Language Models are transformer-based models trained on massive text corpora, capable of performing a wide range of language tasks.

■ GPT-4 by OpenAI, released in March 2023, is a multimodal LLM capable of processing both text and images.

■ LLaMA (Large Language Model Meta AI) was released by Meta AI in February 2023.

■ Claude, developed by Anthropic, is a family of AI assistants focused on safety and helpfulness.

■ Gemini, developed by Google DeepMind, was announced in December 2023 and is natively multimodal.

■ Mistral 7B, released by Mistral AI in September 2023, achieved performance competitive with larger models.

■ Retrieval-Augmented Generation (RAG) combines a retrieval system with a generative model to ground responses in external knowledge.

■ RAG was introduced by Lewis et al. from Facebook AI Research in a 2020 paper titled 'Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks'.

■ Vector databases such as Pinecone, Weaviate, Chroma, and FAISS are commonly used as the retrieval backbone in RAG systems.

■ FAISS (Facebook AI Similarity Search) is an open-source library for efficient similarity search and clustering of dense vectors.

8.3 Benchmarks

■ ImageNet is a large visual database with over 14 million images used for training and benchmarking computer vision models.

■ GLUE (General Language Understanding Evaluation) is a benchmark for evaluating NLP models across multiple tasks.

■ SuperGLUE is a more challenging successor to GLUE, introduced in 2019.

■ MMLU (Massive Multitask Language Understanding) evaluates LLMs across 57 academic subjects.

■ HumanEval, introduced by OpenAI, evaluates code generation capabilities of LLMs using programming problems.

Note: This document was generated specifically for RAG application testing. All factual statements are intended to serve as ground-truth entries for evaluating retrieval accuracy, answer correctness, and context utilisation.